

Constrained Reinforcement Learning with Cumulative or Instantaneous Constraints

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Overview

- 1 Problem Statements of (PC) and (PI)
- 2 Strong duality holds for (PC)
- 3 Instantaneous constrained problems (PI)

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Cumulatively Constrained RL

Let $t \in \mathbb{N} \cup \{0\}$ denote the time instant and $S \subset \mathbb{R}^n$, $A \subset \mathbb{R}^d$ denote states and actions of an agent described by a Markovian dynamical system with transition density $p : s_{t+1} \sim p(\cdot | s_t, a_t)$. The agent chooses actions sequentially based on a policy $\pi \in \mathcal{P}(S)$, space of probability measures.

The goal of (cumulatively) constrained RL (CRL) is to find a policy $\pi^* \in \mathcal{P}(S)$ that meets these objectives by solving the following problem:

$$\begin{aligned} \max_{\pi} \quad & V_0(\pi) \triangleq \mathbb{E}_{s, \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_0(s_t, \pi(s_t)) \right] \\ \text{subject to} \quad & V_i(\pi) \triangleq \mathbb{E}_{s, \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_i(s_t, \pi(s_t)) \right] \geq c_i, i \in \{1, 2, \dots, m\}. \end{aligned} \quad (\text{PC})$$

If one removes the constraints, (PC) degenerates back to "usual" RL formulation, which can be tackled by some standard RL algorithms.

We'll soon spend some time to see that (PC) can be "solved" by primal-dual methods.

Instantaneously Constrained RL

The goal of instantaneously constrained RL is to find a policy $\pi^* \in \mathcal{P}(S)$ that meets these objectives by solving the following problem:

$$\begin{aligned} \max_{\pi} \quad & V_0(\pi) \triangleq \mathbb{E}_{s,\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_0(s_t, \pi(s_t)) \right] \\ \text{subject to} \quad & \mathbb{E}_{\pi} [r_i(s_t, \pi(s_t))] \geq 0, \quad \forall t \in \{0, 1, \dots\}, i \in \{1, 2, \dots, m\}. \end{aligned} \quad (\text{PI})$$

Note that the constraints are applied for **every time instant** t , rather than a long-term weighted sum.

We remark here that an optimal policy feasible for (PI) could be a conservative but feasible solution for (1).

If the policy π is deterministic, then the constraints will become:

$$r_i(s_t, \pi(s_t)) \geq 0, \quad \forall t \in \{0, 1, \dots\}, \forall i \in \{1, 2, \dots, m\}. \quad (1)$$

We will see why (PI) cannot be easily solved under the framework of (PC).

I want to make the slides self-contained. So I include some proofs. I tried to make them as simple as they should be.

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Strong duality proof p1: an important lemma

Lemma (no need for $V_0(\cdot), \dots$ convex.)

If Slater's condition holds for (PC) and its perturbation function $P(\xi)$ is concave, then strong duality holds for (PC).

$$\begin{aligned} P(\xi) &\triangleq \max_{\pi} V_0(\pi) \\ \text{s.t. } &V_i(\pi) \geq c_i + \xi_i, \quad i = 1, \dots, m. \end{aligned}$$

Proof.

First, note that $P(\mathbf{0}) = P^*$. Our goal is $P^* = D^*$. Weak duality always holds:
 $P^* \leq D^* \triangleq \min_{\lambda \geq 0} d(\lambda) = \min_{\lambda \geq 0} \max_{\pi} V_0(\pi) + \sum_{i=1}^m \lambda_i (V_i(\pi) - c_i).$

Slater's condition implies that $\mathbf{0} \in \text{interior of domain of pert. problem } P(\cdot)$. As $P(\xi)$ is concave in ξ , $\exists \lambda \in \mathbb{R}^m$ s.t. $\forall \xi, P(\xi) \leq P(\mathbf{0}) + \lambda^T \xi$. (superdifferential)

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 Further note that those $\lambda \leq 0$. Otherwise, pick $\lambda_j > 0$ and $\xi_j < 0$, ($\xi_i = 0, \forall i \neq j$), we have the following contradiction:

$$P(\mathbf{0}) \leq P(\xi) \leq P(\mathbf{0}) + \lambda^T \xi < P(\mathbf{0}).$$

Then, $\mu = -\lambda \geq 0$, $P^* \geq \sup_{\xi} P(\xi) + \mu^T \xi = \sup_{\pi, \xi: V(\pi) \geq c + \xi} V_0(\pi) + \mu^T \xi =$
 $\sup_{\pi} V_0(\pi) + \mu^T (V(\pi) - c) = d(\mu)$. Thus $D^* = \min_{\mu'} d(\mu') \leq d(\mu) \leq P^*$. □

Strong duality proof p2: $P(\xi)$ is concave

Transform (via some tedious but standard algebra) the perturbed problem into a **linear program** over **occupation measures** $\rho(s, a) \triangleq (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t p_{\pi}^t(s, a)$:

$$P(\xi) = \max_{\pi \in \mathcal{P}(S)} V_0(\pi) = \max_{\rho \in \mathcal{R}} \frac{1}{1 - \gamma} \int r_0(s, a) \rho(s, a) ds da$$

$$\text{s.t. } V_i(\pi) = \frac{1}{1 - \gamma} \int r_i(s, a) \rho(s, a) ds da \geq c_i + \xi_i.$$

where $\mathcal{R}$ is the set of all valid $\rho(\cdot, \cdot)$'s, the discounted state-action occupancies. A crucial property (from Borkar '88) is: $\mathcal{R}$ is a **convex and compact** set.

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Concavity follows: For two perturbations ξ^1, ξ^2 , let ρ_1, ρ_2 be their underlying optimal occupancies. For any $\mu \in (0, 1)$, the convex combination $\rho_{\mu} = \mu \rho_1 + (1 - \mu) \rho_2$ exists (via **convexity** mentioned above, denote its policy as π_{μ}), and is feasible for $\mu \xi^1 + (1 - \mu) \xi^2$ because constraints are linear in ρ . Thus,

$$P(\mu \xi^1 + (1 - \mu) \xi^2) \geq V_0(\pi_{\mu}) = \mu V_0(\pi_1) + (1 - \mu) V_0(\pi_2) = \mu P(\xi^1) + (1 - \mu) P(\xi^2).$$

The above inequality's head and tail match the definition of concavity.

Parametrization does not harm (much)

The previous two slides established strong duality of (PC), which is good from a theoretical point of view. In practice, we may use a **parametrized policy** π_θ (e.g., neural network) with $\theta \in \mathbb{R}^p$ denotes the network weights, etc.. This turns (PC) into (PPC):

$$P_\theta^* \triangleq \max_{\theta} V_0(\theta) \quad \text{s.t.} \quad V_i(\theta) \geq c_i, \quad i = 1, \dots, m. \quad (\text{PPC})$$

How much performance do we lose by using π_θ instead of $\mathcal{P}(S)$?

Definition (ϵ -universal approximator)

A parametrization π_θ is an ϵ -universal parametrization of functions in $\mathcal{P}(S)$ if $\forall \pi \in \mathcal{P}(S)$, we have a π_θ of it: $\max_{s \in S} \int_{\mathcal{A}} |\pi(a|s) - \pi_\theta(a|s)| da < \epsilon$.

If π_θ is an ϵ -universal approximator and rewards are bounded by B_r , then $P^* \geq D_\theta^* \geq P^* - (B_{r_0} + \|\lambda_\epsilon^*\|_1 B_r) \frac{\epsilon}{1-\gamma}$. where D_θ^* is the dual optimum of the parametrized problem, and $\lambda_\epsilon^* \geq 0$ denotes the optimal dual solution to $P(\frac{\epsilon B_r}{1-\gamma})$.

Interpretation: The gap is linear in ϵ . By choosing a richer parametrization (e.g., larger neural network) $\epsilon \rightarrow 0$, the gap vanishes. ☺

The proposed algorithm for (PPC), (PC)

Algorithm 1 dualDescent

Input: η

- 1: *Initialize:* $\theta_0 = 0, \lambda_0 = 0$
 - 2: **for** $k = 0, 1 \dots$
 - 3: Compute an approximation of $\theta_{k+1} \approx \operatorname{argmax} \mathcal{L}_\theta(\theta, \lambda_k)$ with a RL algorithm
 - 4: Compute the dual ascent step $\lambda_{k+1} = [\lambda_k - \eta (V(\theta_{k+1}) - s)]_+$.
 - 5: **end**
-

Remark

- Line 3 could be itself a very slow subroutine.
- According to last slides, a too poor parametrization could make the entire problem infeasible. ($D_\theta^* = -\infty$, λ_ϵ^* has infinite norm,...)

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The "clipping" trick of (PI)

Let's move to problems with instantaneous constraints (PI):

$$\begin{aligned} \max_{\pi} \quad & V_0(\pi) \triangleq \mathbb{E}_{s,\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_0(s_t, \pi(s_t)) \right] \\ \text{subject to} \quad & \mathbb{E}_{\pi} [r_i(s_t, \pi(s_t))] \leq 0, \quad \forall t \in \{0, 1, \dots\}, i \in \{1, 2, \dots, m\}. \end{aligned} \quad (\text{PI})$$

Again, an extremely important assumption is: **Slater condition holds**. That is: (PI)'s feasible region has a nonempty relative interior.

Then a trick called "clipping" could be applied to (PI). Namely: if $r_i = \text{ReLU}(h_i)$ for some function h_i :

$$\begin{aligned} \mathbb{E}_{\pi} [\text{ReLU}(h_i(s_t, \pi(s_t)))] \leq 0 &\iff \mathbb{E}_{\pi} [r_i(s_t, \pi(s_t))] \leq 0 \\ \iff \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t \underbrace{\text{ReLU}(h_i(s_t, \pi(s_t)))}_{\text{treat as new constraints}} \right] &\leq 0. \end{aligned}$$

Then, such (PI) can be cast to the previous (PC) framework... Emm, in fact, not really. The above may work, but could converge very slowly (later we see figures).

A practical surrogate function

Inspired by augmented Lagrangian method (see e.g., Chapter 17 of ¹), consider the following time-invariant instantaneous reward:

$$\tilde{r}(s_t, a_t) \triangleq r_0(s_t, a_t) - \sum_{i=1}^m \lambda_i \text{ReLU}(r_i(s_t, a_t)) - \frac{\rho}{2} \sum_{i=1}^m \text{ReLU}(r_i(s_t, a_t))^2. \quad (2)$$

Denote $R(\pi, \lambda, \rho) \triangleq \mathbb{E}_{\pi} [\sum_{t=0}^{\infty} \gamma^t \tilde{r}(s_t, \pi(s_t))]$.

Then, the algorithm is proposed as the following:

Algorithm 2: Augmented Lagrangian RL

Pick $c_{\rho} \in [1, \infty)$, dual ascent stepsize $\ell \in \mathbb{R}_+$, and convergence tolerance $\varepsilon > 0$;

Initialize $\rho^{(0)} \in \mathbb{R}_+$, $\lambda^{(0)} = 0$;

Randomly initialize the policy π_0 ;

for $k = 0, 1, 2, \dots$ **do**

$\pi_k \leftarrow \arg \max_{\pi} R(\lambda^{(k)}, \rho^{(k)}, \pi)$;

$\lambda^{(k+1)} \leftarrow \left[\lambda^{(k)} - \ell \left(\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \text{Relu}(r(z_t, a_t)) | \pi_k \right] \right) \right]_-$

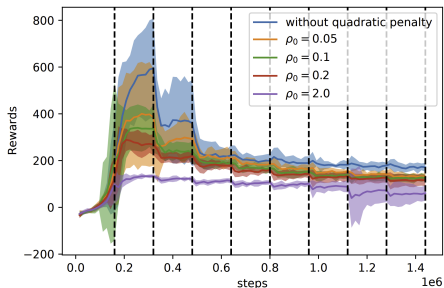
$\rho^{(k+1)} \leftarrow c_{\rho} \rho^{(k)}$;

return π_k if $\|\pi_k - \pi_{k-1}\|_{\infty} \leq \varepsilon$.

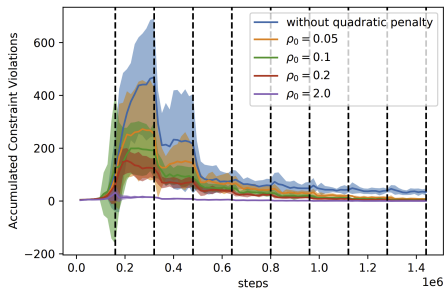
constraint function(s)

¹JS06 Numerical Optimization

A detailed figure



(a) rewards vs. steps



(b) constraint violations vs. steps

Some explanations

- "Without quadratic" can gradually improve. But it seems to take more time.
- Increasing $\rho > 0$ will let the agent to eliminate violations more quickly. But a larger ρ will lead to possibly lower reward value (more conservative).

Some remarks

Rigorous instantaneous feasibility is hard to prove. Because there may be a policy π_{in} being cumulatively feasible, and **instantaneously infeasible** (nearly feasible), meanwhile π_{in} leads to a great reward. Then π_{in} may be returned by the algorithm.

If there exists a sequence $\{\pi_j\}_j$ of infeasible policy which instantaneous infeasibility of π_j being $\frac{1}{j}$. Then there is no $\rho < \infty$ that can rule out all infeasible policies.

To eliminate such problem, further assumptions are needed. E.g., the problem is finite...

Thank You!